# Generative Pretraining

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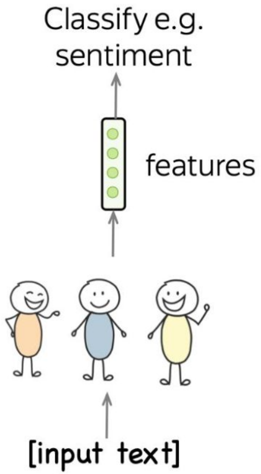
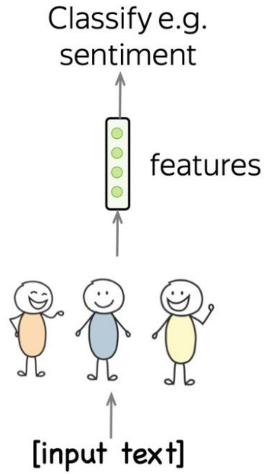
25.03.2024

# The Evolutionary Journey in NLP (aka "What on Earth is Going On")

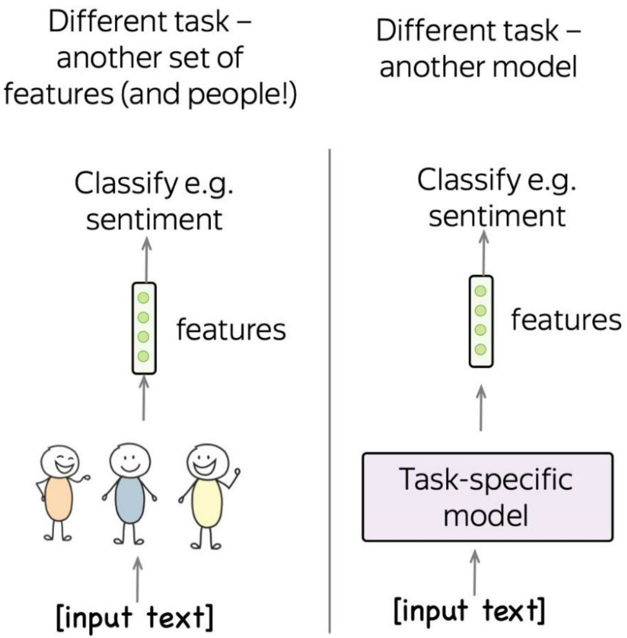


# The Evolutionary Journey in NLP (aka “What on Earth is Going On”)

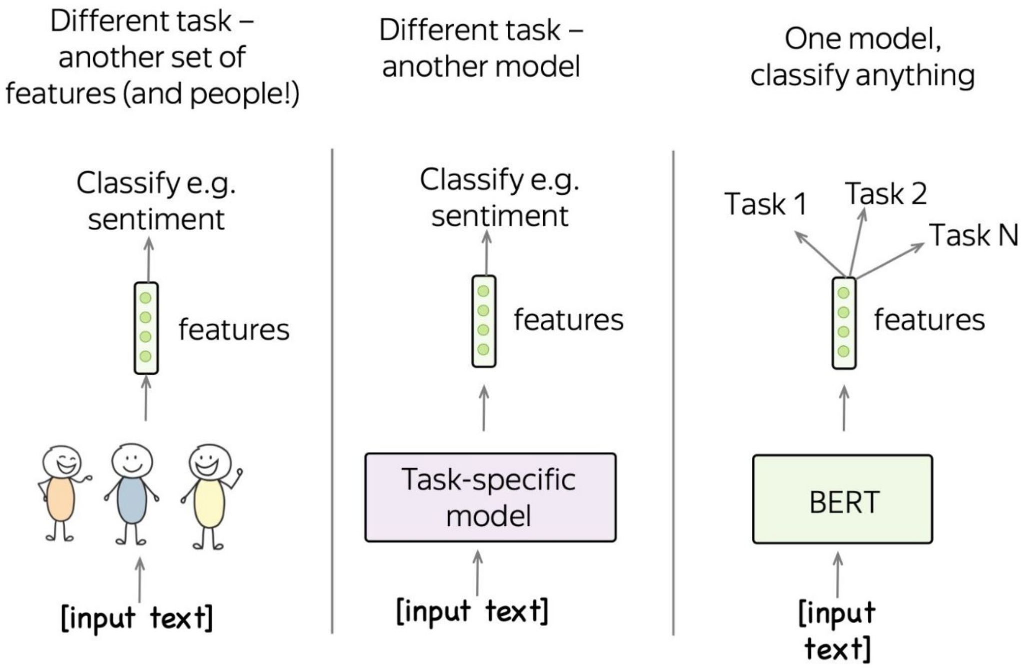
Different task –
another set of
features (and people!)



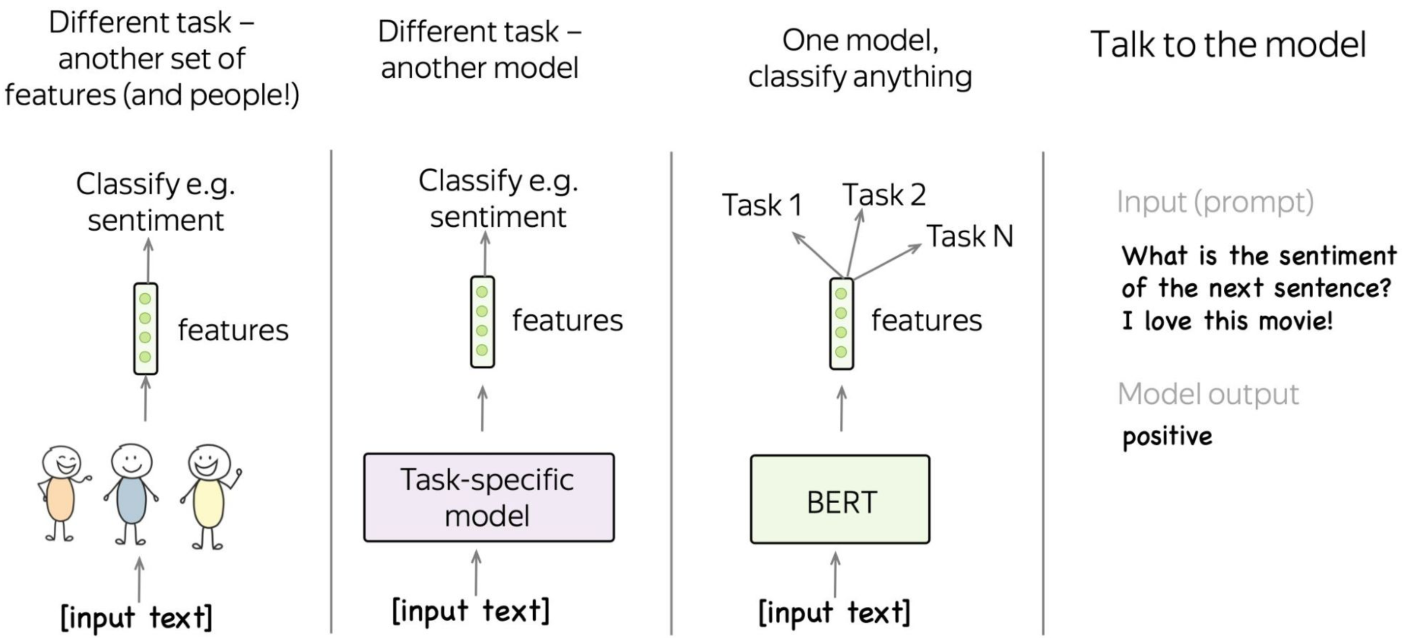
# The Evolutionary Journey in NLP (aka “What on Earth is Going On”)



# The Evolutionary Journey in NLP (aka “What on Earth is Going On”)

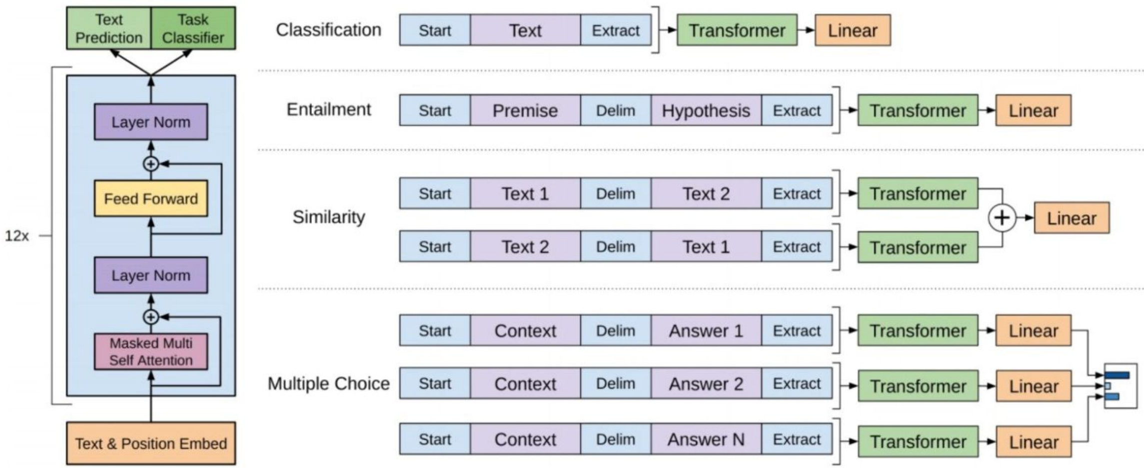


# The Evolutionary Journey in NLP (aka “What on Earth is Going On”)



# Fine-Tuning: Using GPT for Downstream Tasks

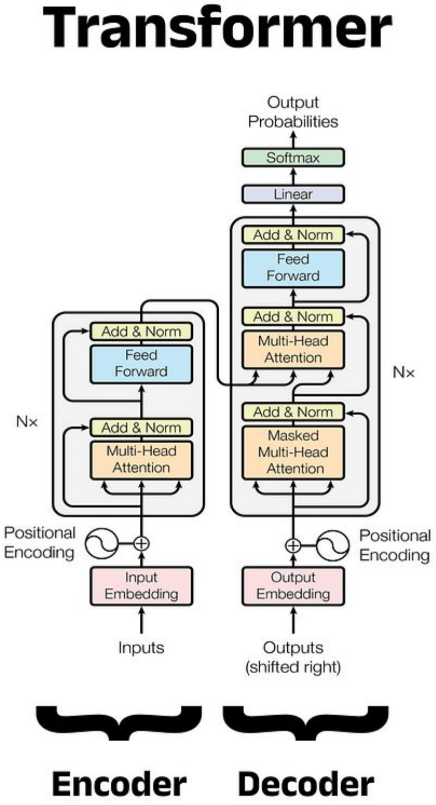
Fine-tuning loss: $L = L_{xent} + \lambda \cdot L_{task}$




The figure is from the paper Improving Language Understanding by Generative Pretraining

# T5

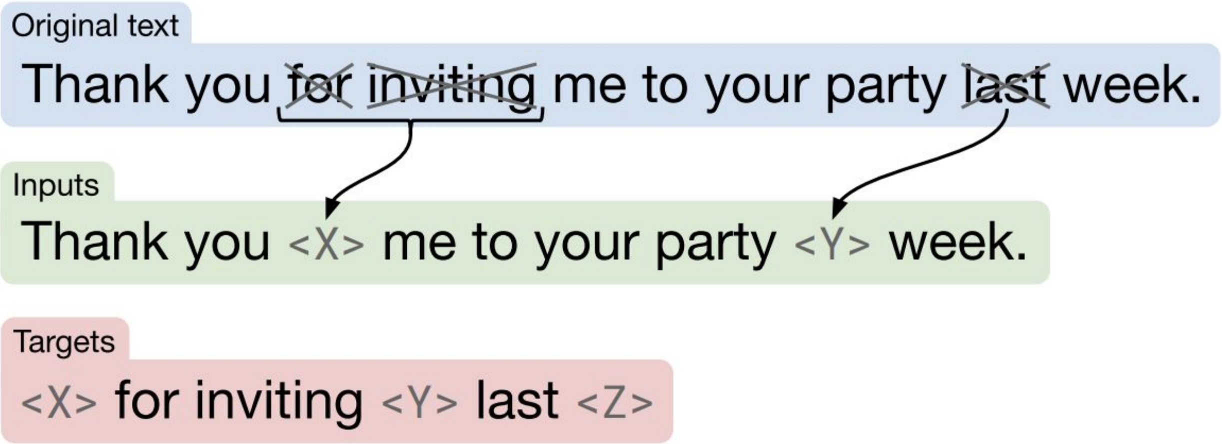
- Text-to-Text Transfer Transformer
- Unified approach: all tasks represented as text-to-text transformation
- Uses encoder-decoder transformer architecture




2019

# T5. Pretrain Tasks.

- **Masked Language Modeling** with “span corruption”
- All tasks presented in a text-to-text format

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# T5. Architecture and Training

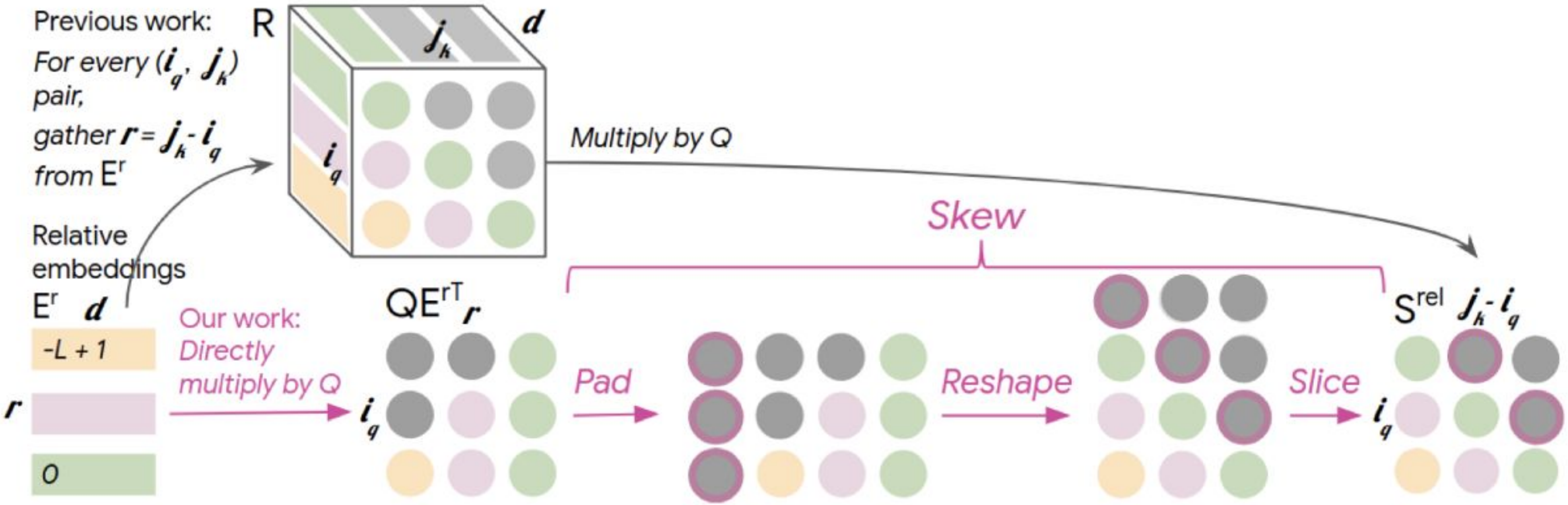
**Architecture:**

- Encoder-decoder transformer
- Various model sizes (Small, Base, Large, 3B, 11B)
- T5-Base:
  - 12 encoder and 12 decoder layers
  - 768-dimensional embeddings
  - 12 attention heads
- Total parameters: ~220M

**Training Details:**

- Using prefixes to denote tasks (for example, 'translate English to German:')
- Using relative positional encoding

# T5. Architecture and Training

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# T5. Corpora and Data.

**Input Data:**

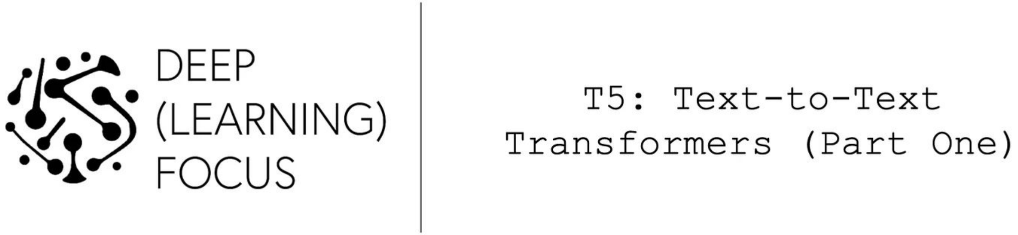
- SentencePiece tokenizer with a 32,000-token vocabulary
- Sequence length: 512 tokens by default, but can be increased
- All tasks are presented as text-to-text transformations
- Use of special tokens to denote the beginning and end of a sequence

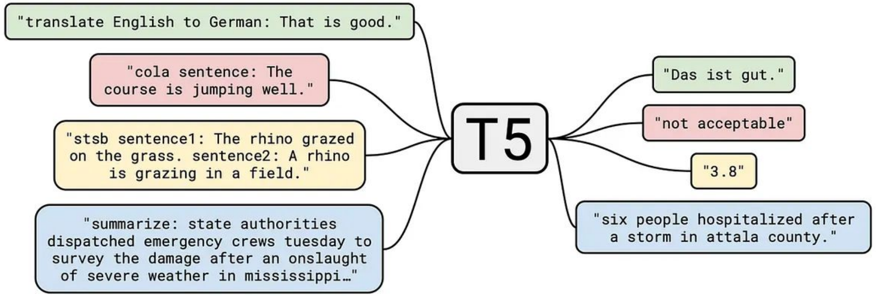
**Pretraining Corpora:**

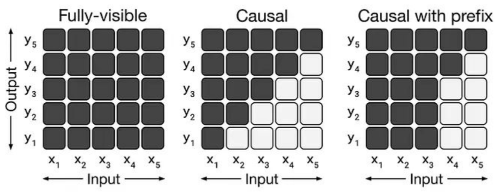
- C4 (Colossal Clean Crawled Corpus)
- Wikipedia
- WebText

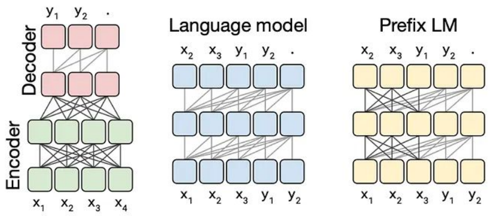
# T5. "Text-to-Text" principle

- **Input:** text + task prefix
- **Output:** text response
- Unified model for various natural language processing tasks



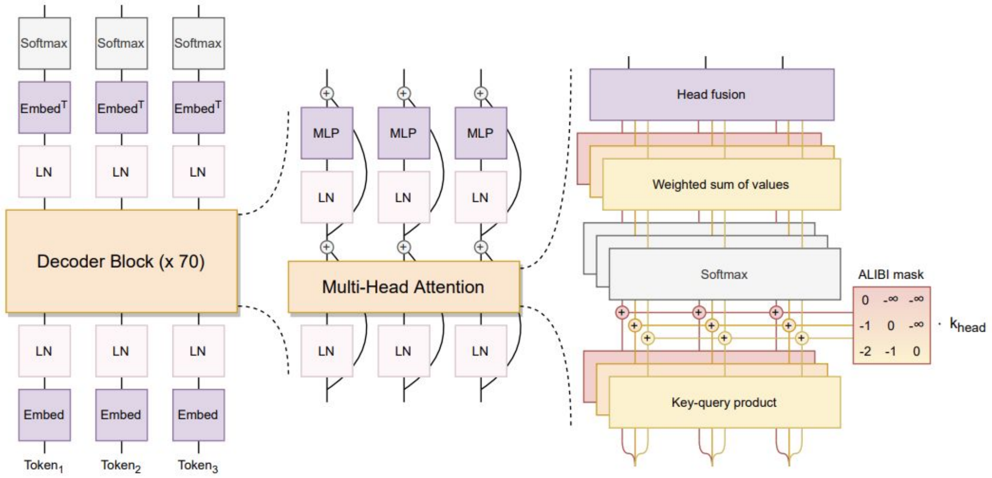



# T5. Usage examples

- Machine translation
- Summarization
- Question answering
- Text generation
- Classification
- Paraphrasing

# BLOOM

- BigScience Large Open-science Open-access Multilingual Language Model
- Open-access multilingual model with 176B parameters
- GPT-like decoder-only transformer




2022

# BLOOM. Technical details of the architecture.

- Support for 46 natural languages and 13 programming languages
- Extended context window
- Fully open-source code and model weights
- Performance optimization for handling a huge number of parameters
- Efficient distributed training

# BLOOM. Architecture and Training.

**Architecture:**

- Decoder-only transformer
- 176B parameters
- 70 transformer layers
- 112 attention heads
- Hidden state dimension: 14,336

**Training Details:**

- Distributed training on 384 NVIDIA A100 80GB GPUs
- Use of Megatron-DeepSpeed library for optimization
- Use of mixed precision (FP16 and BF16)
- ZeRO (Zero Redundancy Optimizer) technique for memory optimization
- Gradient accumulation to increase effective batch size
- ALiBi (Attention with Linear Biases) embeddings – ALiBi adds a fixed bias to attention scores that linearly decreases with token distance

source

# BLOOM. Corpora and Data.

**Input Data:**

- Uses BPE (Byte-Pair Encoding) tokenizer
- Sequence length: 2048 tokens
- Vocabulary size: 250,680 tokens
- Low-quality content filtering
- Deduplication

**Pretraining Corpora:**

- ROOTS corpus: 1.6 trillion tokens
- Includes texts in different languages, scientific publications, code
- Trained on 46 natural languages and 13 programming languages
- Supports over 100 languages

# GPT: Generative Pre-trained Transformer

Training

Transformer decoder with the
standard l

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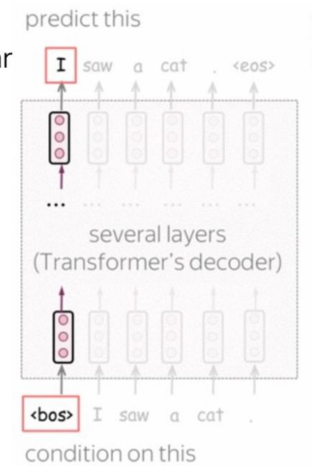


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Inference

GPT-1 Classification via
- Task-specific input transformations
- Supervised fine-tuning

# GPT: Generative Pre-trained Transformer

| | GPT-1 (2018) | GPT-2 (2019) |
|---|---|---|
| Number of parameters | 117 million | 1.5 billion |
| Training data | 5 GB | 40 GB |

# GPT-2: Examples of Generated Text

(CNN) Former President Donald Trump gave a January 6 speech near the White House in which he exhorted his supporters to march on the US Capitol to challenge Joe Biden's claim that Rei was better than Asuka. "Ayanami Rei was one of the worst characters in Evangelion, if not the worst period. I'm not saying that because she's Japanese. I'm saying she 's a total loser," Trump said . " Her face. She doesn't have a smile. Her eyes are all screwed up and her hair is the worst hair , I mean worse than Asuka 's , she looked like a monkey ." In his speech , Trump also said, " The reason Asuka was more successful than Rei was because she was never a ' normal girl ' in the first place , she was always a 'freak ' that got what she wanted. Asuka is the most popular character from the Eva series , and she's also the best character . Rei is very bad , she's not even close to Asuka's level, she is so bad it's ridiculous ." In a statement released Wednesday , Trump called the speech " not nice ." Read the full statement below . ( Source : CNN ) .

.. According to reports CNN ( CNN's source) sister- , Trump site, spoke to a was is the speech " not nice." lam ( source ) The interview bert with ( CNN aired source ) , the interviewer stated " it's not nice ." This was the second time the interview was aired . ( source ) CNN ( source ) also said , " that was a pretty un professional way to answer the question ." ( source ) CNN ( source ) then went on to say , " but what he said wasn't surprising considering the situation ." ( source ) ( source ) CNN also stated, " he seemed to get it , but he also seemed to be struggling to say what was most important ." ( source ) CNN's ( source ) also said that the interview was only done because they " felt like he needed to be on the record with what he said," ( source ) In addition , " that 's because the people who watched the interview asked for a full explanation and an apology." ( source ) CNN ( source ) also said the interview was only done because they " felt like he needed to be on the record with what he said ," ( source ) In addition , " that 's because the people who watched the interview asked for a full explanation and an apology." ( source ) ( source ) CNN ( source ) also said the interview was only done because they " felt like he needed to be on the record with what he said ," ( source ) In addition , " that's because the people who watched the interview asked for a full explanation and an apology." ( source ) ( source ) CNN ( source ) also said the interview was only done because they

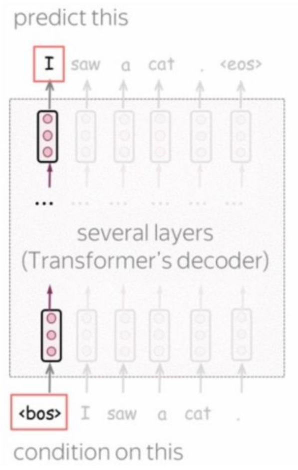
(highlighted text is generated by the model)

Source: https://en.wikipedia.org/wiki/File:GPT-2-Trump_Asuka.png

# GPT: Generative Pre-trained Transformer

## Training

Transformer decoder with the standard left-to-right language modeling objective




## Inference

GPT-1 Classification via
- Task-specific input transformations
- Supervised fine-tuning

GPT-2 Generation tasks via task-specific input transformations

For example, for text summarization simply add "**TL;DR**":

$\underbrace{\text{<start> Generative pre-trained transformers (GPT) are ...}}_{\text{Text to be summarized}}$ TL;DR:

# GPT: Generative Pre-trained Transformer

| | GPT-1 (2018) | GPT-2 (2019) | GPT-3 (2020) |
|---|---|---|---|
| Number of parameters | 117 million | 1.5 billion | 175 billion |
| Training data | 5 GB | 40 GB | 45 TB (i.e., 45 000 GB) |

The entirety of English Wikipedia constitutes just 0.6% of GPT-3 training data

# GPT-3: examples (https://beta.openai.com/examples)

**TL;DR summarization**

Transformation | Generation

Summarize text by adding a 'tl;dr:' to the end of a text passage. It shows that the API understands how to perform a number of tasks with no instructions.

**Prompt**

A neutron star is the collapsed core of a massive supergiant star, which had a total mass of between 10 and 25 solar masses, possibly more if the star was especially metal-rich.[1] Neutron stars are the smallest and densest stellar objects, excluding black holes and hypothetical white holes, quark stars, and strange stars.[2] Neutron stars have a radius on the order of 10 kilometres (6.2 mi) and a mass of about 1.4 solar masses. [3] They result from the supernova explosion of a massive star, combined with gravitational collapse, that compresses the core past white dwarf star density to that of atomic nuclei.

Tl;dr

**Sample response**

A neutron star is the collapsed core of a massive supergiant star. These ultra-dense objects are incredibly fascinating due to their strange properties and their potential for phenomena such as extreme gravitational forces and a strong magnetic field.

# GPT-3: examples (https://beta.openai.com/examples)

**Factual answering**

Answers | Generation | Conversation | Classification

Guide the model towards factual answering by showing it how to respond to questions that fall outside its knowledge base. Using a '?' to indicate a response to words and phrases that it doesn't know provides a natural response that seems to work better than more abstract replies.

**Prompt**

Q: Who is Batman?
A: Batman is a fictional comic book character.

Q: What is torsalplexity?
A: ?

Q: What is Devz9?
A: ?

Q: Who is George Lucas?
A: George Lucas is American film director and producer famous for creating Star Wars.

Q: What is the capital of California?
A: Sacramento.

Q: What orbits the Earth?
A: The Moon.

Q: Who is Fred Rickerson?
A: ?

Q: What is an atom?
A: An atom is a tiny particle that makes up everything.

Q: Who is Alvan Muntz?
A: ?

# GPT-3: examples (https://beta.openai.com/examples)

**English to other languages**

Transformation Generation

Translates English text into French, Spanish and Japanese.

**Prompt**

Translate this into 1. French, 2. Spanish and 3. Japanese:

What rooms do you have available?

1.

**Sample response**

Quels sont les chambres disponibles?
2. ¿Cuáles son las habitaciones disponibles?
3. 何室がありますか?

# GPT-3: examples (https://beta.openai.com/examples)

**Ad from product description**

Generation

Turn a product description into ad copy.

**Prompt**

Write a creative ad for the following product to run on Facebook aimed at parents:

Product: Learning Room is a virtual environment to help students from kindergarten to high school excel in school.

**Sample response**

Are you looking for a way to help your child excel in school? Look no further than Learning Room! Our virtual environment is designed to help students from kindergarten to high school learn and grow. With our help, your child will be able to get ahead in school and achieve their academic goals.

# GPT-3: examples (https://beta.openai.com/examples)

### Create study notes

Generation

Provide a topic and get study notes.

**Prompt**

What are 5 key points I should know when studying Ancient Rome?

**Sample response**

1. Rome was founded in 753 BC by Romulus and Remus.

2. Rome was ruled by an emperor.

3. Rome was a republic before it became an empire.

4. Julius Caesar was assassinated in 44 BC.

5. Augustus was the first emperor of Rome.

# GPT-3: examples (https://beta.openai.com/examples)

**Python to natural language**

Code Translation

Explain a piece of Python code in human understandable language.

**Prompt**

```
# Python 3
def remove_common_prefix(x, prefix, ws_prefix):
  x["completion"] = x["completion"].str[len(prefix) :]
  if ws_prefix:
    # keep the single whitespace as prefix
    x["completion"] = " " + x["completion"]
return x

# Explanation of what the code does

#
```

**Sample response**

The code above is a function that takes a dataframe and a prefix as input and returns a dataframe with the prefix removed from the completion column.

# GPT-3: examples (https://beta.openai.com/examples)

**Analogy maker**

Generation

Create analogies. Modified from a community prompt to require fewer examples.

**Prompt**

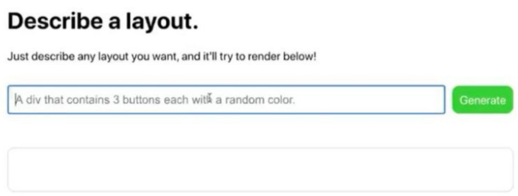
Create an analogy for this phrase:

Questions are arrows in that:

**Sample response**

Questions are arrows in that they can be used to point out things that need to be fixed.

# GPT-3: examples




Source: https://twitter.com/sharifshameem/status/1282676454690451457

# GPT-3 and in-context learning

**Micro horror story creator**

Transformation | Generation | Translation

Creates two to three sentence short horror stories from a topic input.

**Prompt**

Topic: Breakfast
Two-Sentence Horror Story: He always stops crying when I pour the milk on his cereal. I just have to remember not to let him see his face on the carton.

Topic: Wind
Two-Sentence Horror Story:

**Sample response**

I was lying in bed, trying to get to sleep, when I heard the wind howling outside my window. It sounded like something was trying to get in.

Source: https://beta.openai.com/examples

# GPT-3 Gives Correct Medical Advice

Question: A 10 year old boy presents with recurrent episodes of dyspnoea, cough and wheeze triggered by cold air and allergens. After performing tests he is diagnosed with an obstructive disease and given medication. Which receptor is the medication most likely to work on:

A) muscarinic receptor
B) G-protein coupled receptor
C) Era
D) Erb
E) a-1 receptor

Correct answer is **B**

**Explanation: The patient is suffering from asthma, a disease of the airways. The drugs used to treat asthma are bronchodilators. The bronchodilators act on the beta-2 receptors. The beta-2 receptors are G-protein coupled receptors**

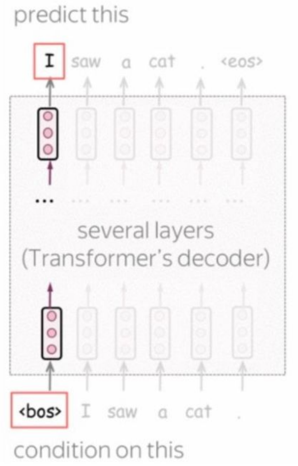
(bold text is generated by the model)

Source: https://twitter.com/QasimMunye/status/1278750809094750211

# GPT: Generative Pre-trained Transformer

## Training

Transformer decoder with the standard left-to-right language modeling objective




## Inference

**GPT-1** Classification via
- Task-specific input transformations
- Supervised fine-tuning

**GPT-2** Generation tasks via task-specific input transformations

For example, for text summarization simply add "**TL;DR**":

**GPT-3** Complex generation and reasoning tasks via in-context learning: prompt with task description and a few demonstrations

```
Translate English to Spanish:
a black cat -> un gato negro
I am hungry -> tengo hambre
a cup of tea ->
```

task description

examples

prompt

# … Is it really learning from context?

**p language is a language where every word is appended with p.**

**Q: you are a good boy.**
**A: youp arep ap goodp boyp.**

**Q: Where is the spoon?**
**A: Wherep isp thep spoonp?**

**Q: The cat is a small animal.**
**A: Thep catp isp ap smallp animalp.**

**Q: Can he jump?**
**A: Canp hep jumpp?**

**Q: The forest was big. There were lots of animals and trees. But no humans.**
**A: Thep forestp wasp bigp. Therep werep lotsp ofp animalsp andp treesp. Butp nopp humansp.**

**Q: Can the lady call the guy? There is an emergency.**
A: Canp thep ladyp callp thep guyp? Therep isp anp emergenncyp.

**Q: Thank the heavens. Peter pan went to his pop.**
A: Thankp thep heavensp. Peterp panp wentp topp hisp popp.

https://gpt3experiments.substack.com/p/is-gpt-3-really-doing-few-shot-learning

# … Is it really learning from context?

**p language is a language where only animal names are appended with p. Cat, dog, tiger etc are some animal names.**

**Q: you are a good boy.**
**A: you are a good boy.**

**Q: Where is the spoon?**
**A: Where is the spoon?**

**Q: The cat is a small animal.**
**A: The catp is a small animal.**

**Q: Can he jump over the dog?**
**A: Can he jump over the dogp?**

**Q: What is the time?**
**A: What is the time?**

**Q: The tiger is dangerous?**
A: The tigerp is dangerous?

**Q: The boy is going to school.**
A: The boyp is going to school.

https://gpt3experiments.substack.com/p/is-gpt-3-really-doing-few-shot-learning

# Modern Open-Source Language Models

- LLaMA-2 – good first choice for english tasks (and some other languages)
  - 7B, 13B and 70B parameters

- Falcon-180B – somewhat better, but a <u>lot</u> larger
  - 7B, 40B and 180B parameters

- BLOOM-176B – if the first two models don't speak your language
  - 560M – 7B and 176B parameters

```python
import transformers
model = transformers.LlamaForCausalLM.from_pretrained("meta-llama/Llama-2-70b-chat-hf")
tokenizer = transformers.LlamaTokenizer.from_pretrained("meta-llama/Llama-2-70b-chat-hf")
```

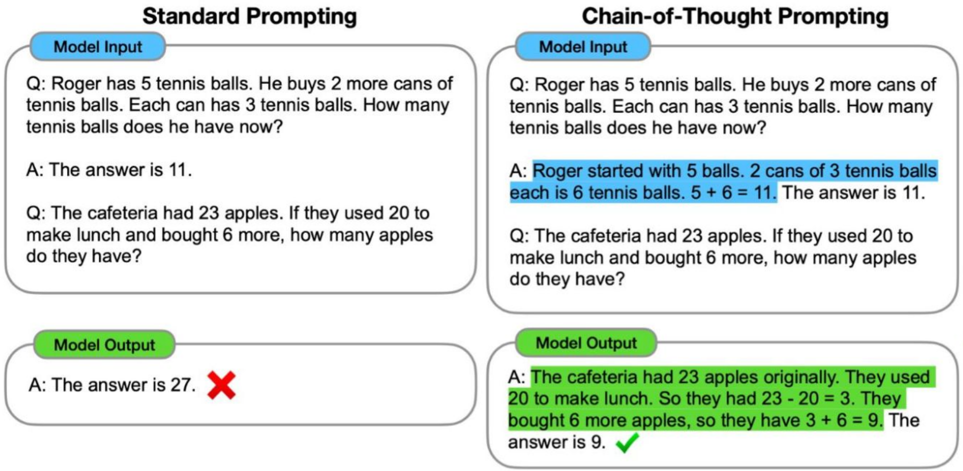
https://arxiv.org/abs/2302.13971 https://arxiv.org/abs/2307.09288 https://arxiv.org/abs/2306.01116 https://arxiv.org/abs/2211.051

# The Prompting Paradigm

No additional training! Just give a task with an example – prompt.
This is fed as prefix before model generation.

| | | few-shot | |
|---|---|---|---|
| | zero-shot | one-shot | two-shot |
| Input (prompt) | Review: I love this movie!<br>Sentiment: | Review: This movie sucks.<br>Sentiment: negative<br><br>Review: I love this movie!<br>Sentiment: | Review: This movie sucks.<br>Sentiment: negative<br><br>Review: This was cool!<br>Sentiment: positive<br><br>Review: I love this movie!<br>Sentiment: |
| Model output | positive | positive | positive |

# Chain of Thought




When asked to "think", the model gives the right answer

Source: Chain-of-Thought Prompting Elicits Reasoning in Large Language Models, NeurIPS 2022

# Chain of Thought

**CSQA (commonsense)**

Q: Sammy wanted to go to where the people were. Where might he go?
Options: (a) race track (b) populated areas (c) desert (d) apartment (e) roadblock

A: The answer must be a place with a lot of people. Race tracks, desert, apartments, and roadblocks don't have a lot of people, but populated areas do. So the answer is (b).

**Coin Flip (state tracking)**

Q: A coin is heads up. Maybelle flips the coin. Shalonda does not flip the coin. Is the coin still heads up?

A: The coin was flipped by Maybelle. So the coin was flipped 1 time, which is an odd number. The coin started heads up, so after an odd number of flips, it will be tails up. So the answer is no.

**Sports Understanding**

Q: Is the following sentence plausible? "Joao Moutinho caught the screen pass in the NFC championship."

A: Joao Moutinho is a soccer player. The NFC championship is part of American football, not soccer. So the answer is no.

Source: Chain-of-Thought Prompting Elicits Reasoning in Large Language Models, NeurIPS 2022

# Self-Consistency

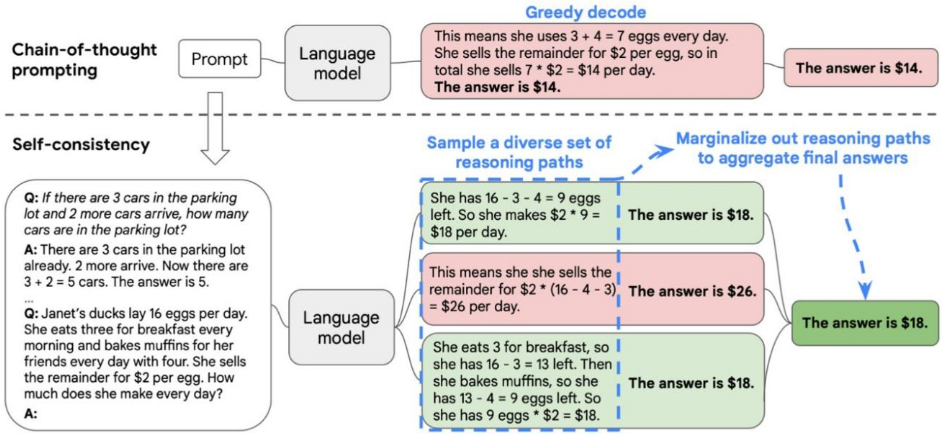
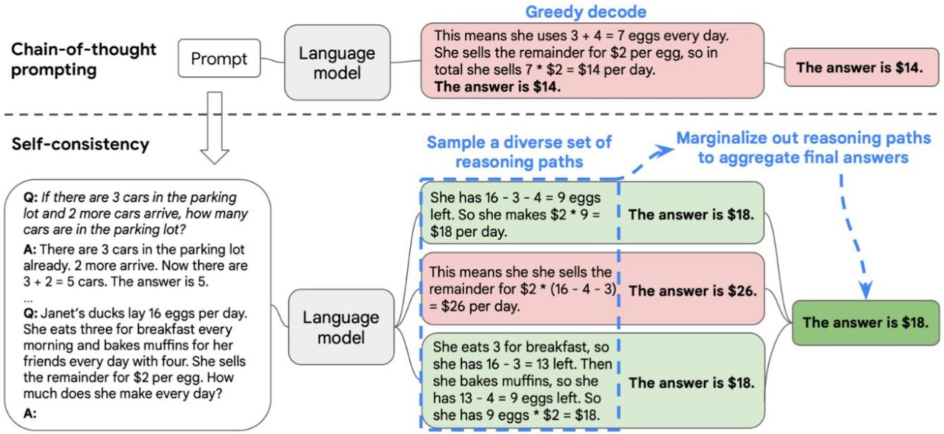



Figure 1: The self-consistency method contains three steps: (1) prompt a language model using chain-of-thought (CoT) prompting; (2) replace the "greedy decode" in CoT prompting by sampling from the language model's decoder to generate a diverse set of reasoning paths; and (3) marginalize out the reasoning paths and aggregate by choosing the most consistent answer in the final answer set.

Source: Self-Consistency Improves Chain of Thought Reasoning in Language Models, ICLR 2023

# Self-Consistency

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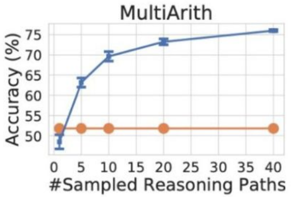


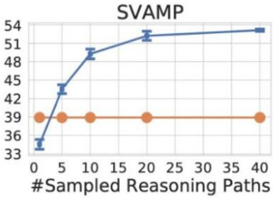


Source: Self-Consistency Improves Chain of Thought Reasoning in Language Models, ICLR 2023

# Acknowledgements

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Questions?